Minseong Kim

Manufacturing IT Engineer · MES · PLM · ERP

Cerritos, CA · medium2000@naver.com · minseong-kim.github.io/career/#manufacturing

SUMMARY

Manufacturing IT engineer with semiconductor roots. At Samsung Electronics I led enterprise PLM (modular and configurable BOM, SAP integration) and the System LSI design-IP management system; later I built warehouse management and inventory sync systems and integrated ERP, WMS and TMS in logistics. Recent MES·AI prototypes model FDC root-cause analysis, lot hold, QMS re-inspection and ECN/BOM coordination between a Korean HQ and a U.S. plant. Studying Electrical Technology – Automation at LBCC.

SKILLS

Databases	Oracle, SQL Server, MySQL, PostgreSQL, MongoDB, Data modeling (ERwin), Query tuning
AI & ML	Claude API (LLM), Tool use, RAG, Random Forest, Isolation Forest, Anomaly detection, Predictive maintenance
Manufacturing & enterprise	MES, PLM, BOM / ECN, ERP & SAP integration, WMS, TMS, QMS, FDC / RCA
Leadership & delivery	Project leadership, Requirements analysis, UAT, Vendor coordination, Technical documentation, Mentoring
Languages	Java, C#, Python, TypeScript, JavaScript, SQL, Rust, Shell Script, Groovy
Front end	React, Next.js, Angular, Vue.js, Material-UI, Tailwind CSS, Kendo UI, HTML5 / CSS3, SPA & SSR

INDEPENDENT AI PROJECTS

MES Process Innovation Console (Solar) — Ingot-to-module MES simulation with AI-assisted root-cause analysis and predictive maintenance using Random Forest and Isolation Forest models.

Battery MES·AI Integration Platform — Six-system cascade (MES, FDC, EMS, RMS, QMS, MCS) with defect classification, anomaly detection and battery-life models; Spring Boot REST APIs and LLM diagnostic explanations.

Switchgear MES·AI Bridge — Bilingual console modeling FDC root-cause analysis, automated HiPot failure response, QMS re-inspection and HQ ↔ plant ECN/BOM coordination.

ClinicalMind – AI-Assisted Clinical Study Design — Three Claude API workflows (draft generation, real-time CDISC coherence check, RAG search of similar protocols) and an immutable audit trail designed for 21 CFR Part 11. Angular 19 architecture patterns.

EXPERIENCE

Independent Technical Projects

08/2025 – Present · Cerritos, CA

MES / AI portfolio (self-directed, non-commercial)

- Built four live demos covering MES/AI integration and LLM-assisted clinical study design (see Independent AI Projects).
- Modeled anomaly detection and predictive maintenance with Random Forest and Isolation Forest on simulated plant data.
- Designed Spring Boot REST and Oracle/SQL Server integration architecture for MES, QMS and material control modules.

FNS Inc

11/2021 – 08/2025 · Torrance, CA

IT Lead Manager (Full-Stack Developer) · global logistics

- Integrated applications with ERP, WMS, TMS, customs systems and third-party logistics providers through RESTful APIs.
- Mentored developers, led technical discussions and coordinated vendors and business teams; wrote deployment procedures and troubleshooting guides.

Nmsi Inc

06/2020 – 10/2021 · Brea, CA

IT Lead Manager (Full-Stack Developer) · mortgage lending

- Led UAT and deployment validation; wrote technical specifications and data flow diagrams.

Very J 11/2019 – 06/2020 · Los Angeles, CA

IT Manager (Full-Stack Developer) · retail and warehouse

- Built a real-time inventory synchronization system that transmitted inventory data between warehouses and retail stores automatically.
- Developed Warehouse Management System (WMS) and Time Card applications with React/Material-UI front ends.

Daegu Bank 05/2016 – 12/2016 · Korea

Project Leader · next-generation banking system

- Designed and developed core banking applications and integrated financial services; led UAT and production implementation.

PV Technology 12/2015 – 05/2016 · Korea

Project Leader · Warehouse Management System

- Led a WMS covering receiving, storage, picking, packing and shipping, including integration, UAT and user training.

Korea Post Information Center 12/2013 – 06/2015 · Korea

Project Leader · Smart Electronic Banking

- Led development of mobile and online banking applications on an integrated mobile banking platform.

Samsung Electronics 06/2011 – 12/2013 · Gyeonggi-do, Korea

Project Leader (Software Engineer) · PLM and System LSI

- Implemented modular, configurable and intelligent BOM in enterprise PLM so multiple end items could be created with less effort and cost.
- Designed a JavaScript client library API that let SAP and legacy applications embed PLM screens.
- Led the System LSI IP management system for cataloging and reusing semiconductor design IP through tree browsing and parametric search.
- Supported overseas users as PLM business administrator and trained power users across divisions.

LG Hitachi — *Software Engineer · financial services* 02/2011 – 06/2011 · Korea

Nexon Mobile — *Software Engineer · mobile web* 09/2010 – 01/2011 · Korea

NTT DATA 01/2007 – 09/2010 · Fukuoka, Japan

Software Engineer · aeronautical information system

- Developed core modules of an Aeronautical Information Publication system supporting high-availability aviation operations.

Neobill — *Software Engineer · electronic billing and payments* 03/2006 – 01/2007 · Korea

Moazio — *Software Engineer · mobile games* 09/2005 – 03/2006 · Korea

EDUCATION

Georgia Institute of Technology 2027

M.S. Computer Science ADMITTED

Long Beach City College 2026 – now

A.S. Electrical Technology – Automation Technician

IN PROGRESS

California University of Management and Sciences 2019

M.S. Computer Information Systems

Jeonju University, Korea 2006

B.E. Information and Communication Engineering · GPA 3.99 / 4.5

CERTIFICATIONS

Oracle Certified Professional (OCP), Database 10g 2010

Sun Certified Business Component Developer (SCBCD) 2010

Sun Certified Web Component Developer (SCWCD) 2010

Sun Certified Java Programmer (SCJP) 2010

JLPT N1 (Japanese, highest level) 2014

Project Management Professional (PMP) 2026

Exam preparation IN PROGRESS

California Electrician Trainee (ET) registration 2026

Languages

Korean (native) · English (professional) · Japanese (business, JLPT N1)

Military service

Republic of Korea Air Force, full term completed